THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–8) | Class 7 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 300	1 crore = 100 lakh.
2	**(d)** 18	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
3	**(c)** 1	Multiply as whole numbers, then count the decimal places in both factors.
4	**(b)** 28	Perimeter $= 4\times$ side.
5	**(b)** 66°	Their sum is 90°.
6	**(c)** 77	Sum $= 11(a+b)$.
7	**(b)** 7 cm	The third side lies between the difference and the sum of the other two.
8	**(b)** 12	Number of pieces $= n \div$ piece length.
9	**(c)** 2	Group digits in threes.
10	**(d)** 23	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
11	**(a)** 8	Multiply both numbers by 10 or 100 so that the divisor becomes a whole number.
12	**(a)** $5n+2$	Write the rule using a letter-number.
13	**(d)** 60°	The angles add up to 180° in 3 equal parts.
14	**(a)** 63	Difference $= 9(a-b)$.
15	**(b)** 30	Add the angles: $(3+k)x = 180$.
16	**(d)** 15	Scaling by a fraction is multiplication.
17	**(b)** 4,560,000	Regroup the digits in threes.
18	**(c)** 7	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
19	**(b)** 345	1 m = 100 cm.
20	**(b)** 16	Combine the like terms first: $m + am = (1+a)m$.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Regroup in threes (1); 2,345,678 (1).	1+1
22	Evaluate inner brackets first; Then the outer operations (1); Value $= 32$ (1).	1+1
23	Multiply without the decimal points: $32 \times 5 = 160$ (1); Decimal places in the factors: $1+1=2$, so the product is 1.6 (1).	1+1
24	Substitute n in $3n+1$ (1); $3(5)+1 = 16$ matchsticks (1).	1+1
25	The angles are y and $2y$, so $3y = 180$; $y = 60$ (1); The angles are 60° and 120° (1).	1+1
26	$10a+b+10b+a = 11a+11b$; $= 11(a+b)$ (1); For $a=3, b=5$: $35+53 = 88 = 11\times 8$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Break the number by place value (1); $3\times10^6 + 4\times10^5 + 5\times10^4 + 6\times10^3 + 7\times10^2 + 8\times10^1 + 9\times10^0$ (1); In lakhs: 34.6 lakh (1).	1+1+1
28	Innermost bracket first; Then outward (1); Then multiply/divide, add/subtract (1); Value $= 28$ (1).	1+1+1
29	Cost $= 125.5 \times 3.5$ (1); Cost $=$ ₹439.25 (2).	1+2
30	Figure 4: $3(4)+2 = 14$ (1); Figure 5: $3(5)+2 = 17$ (1); Difference $= 17-14 = 3$ (1).	1+1+1
31	Fourth angle $= 360^\circ - (95+85+110)^\circ = 70^\circ$ (1); Kinds: 95° obtuse, 85° acute, 110° obtuse, 70° acute (2).	1+2
	OR: Step 1: $9652-2569 = 7083$ (1); Step 2: $8730-0378 = 8352$ (1); After two steps the number is 8352 (1).	1+1+1
32	$\angle B = \angle C$ (equal sides); $(x+10)+2(2x-5) = 180$, so $5x = 180$ (1); $x = 36$ (1); $\angle A = 46^\circ$, $\angle B = \angle C = 67^\circ$ (1).	1+1+1
	OR: Area $= \frac{3}{4}\times\frac{2}{5} = \frac{3}{10}$ m^2 (1); Smaller side squared $= \frac{4}{25}$ m^2 (1); Difference $= \frac{7}{50}$ m^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Annual use $= 1,250 \times 365 = 4,56,250$ litres (1); Nearest lakh: $5,00,000$ (1); In words: five lakh (1); Check by estimation: 365 is about 400 (2).	1+1+1+2
34	Cost price $= 20 \times 80 + 15 \times 120$ (1); $= 3400$ (1); Profit/loss $= 3700 - 3400$ (1); $= 300$ (1); Profit (1).	1+1+1+1+1
35	Rice $= 2.5 \times 48 = 120$ (1); Wheat $= 1.5 \times 36 = 54$ (1); Total $= 174$ (1); Balance $= 200 - 174 = 26$ (2).	1+1+1+2
36	Notebooks cost $12n$ and pens cost 20 (1); Total cost $= 12n + 20$ (1); At $n = 10$: 140 (1); Change $= 200 - 140 = 60$ (2).	1+1+1+2
	OR: Equal to 50°: the angle itself, its vertically opposite angle, its corresponding angle and its alternate interior angle, that is 4 angles (1); Every other angle forms a linear pair with one of these, so it measures 130° (4 angles) (2); Check: $50 + 130 = 180$ (2).	1+2+2
37	Reverse $= 237$ (1); Difference $= 732 - 237 = 495$ (1); $= 99 \times 5$ (1); In general $100a + c - 100c - a = 99(a - c)$ (1); So the difference is always a multiple of 99 (1).	1+1+1+1+1
	OR: $4x + 20 = 180$, so $x = 40$ (1); Angles 50°, 30°, 100° (1); Classification: obtuse-angled (1); Exterior angles 130°, 150°, 80° (1); Their sum $= 360^\circ = 360^\circ$ (1).	1+1+1+1+1
38	First part $= \frac{1}{4} \times 120 = 30$ km (1); Remaining $= 120 - 30 = 90$ km (1); Second part $= \frac{2}{3} \times 90 = 60$ km (1); Left $= 90 - 60$ (1); $= 30$ km (1).	1+1+1+1+1
	OR: People surveyed $= \frac{4}{100} \times 2,50,00,000$ (1); $= 10,00,000$ (2); In words: ten lakh (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–8) | Class 7 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 115°	Interior angles on the same side are supplementary.
2	**(a)** 5	List the terms and count the steps.
3	**(a)** 30	Exterior angle = sum of interior opposite angles.
4	**(a)** 8	Use the meaning of multiplying and dividing fractions.
5	**(b)** 4,00,000	Digit times its place value.
6	**(d)** $(3+4)\times 5-6$	Check which grouping gives the value.
7	**(b)** 2500	1 kg = 1000 g.
8	**(d)** $5x$	Subtract the coefficients of the like terms.
9	**(b)** parallel	Use the tests for parallel lines.
10	**(a)** 2	Digit sum must be a multiple of 9.
11	**(d)** QR	Use the idea of corresponding parts.
12	**(b)** 6	Convert to an improper fraction and multiply.
13	**(c)** 5	Count the zeros.
14	**(d)** 38	Evaluate both.
15	**(c)** 725	₹1 = 100 paise.
16	**(b)** 14	Add inside the bracket first.
17	**(c)** 28°	Vertically opposite angles are equal.
18	**(a)** 55	Triangular number $\frac{n(n+1)}{2}$.
19	**(a)** $\angle A$	The included angle lies between the two given sides.
20	**(c)** $\frac{2}{3}$	Write as an improper fraction, then invert.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Third side $> 6-10$ in absolute value $= 4$; Third side $< 6+10 = 16$ (1); Least whole number 5 cm; Greatest whole number 15 cm (1).	1+1
22	Count $= \frac{3}{4} \div \frac{1}{8}$; $= \frac{3}{4} \times \frac{8}{1} = \frac{24}{4}$ (1); Answer 6 (1).	1+1
23	Successor of 99,999 is 1,00,000 (1); Successor of 99,999 is 1,00,000 (1).	1+1
24	$7 \times 9 + 12$ (1); Value $= 75$ (1).	1+1
25	Make the divisor a whole number: $36 \div 4$ (1); Answer: 9 (1).	1+1
26	Collect like terms: $(7-3)x$ and $(4+6)$ (1); Answer: $4x+10$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	$1,25,000 \times 8 = 10,00,000$ (1); Words: ten lakh (1); International: 1,000,000 (1).	1+1+1
28	Cloth left $= 60 - 12 \times 3$; $= 24$ m (1); Cloth used $= 36$ m (1); Cost $= 36 \times 45 = 1620$ (1).	1+1+1
29	Divide: $3 \div 8 = 0.375$ (1); Compare 0.375 with 0.4 (1); $\frac{3}{8}$ is less than 0.4 (1).	1+1+1
30	Group the x-terms and the y-terms (1); x-terms: $5x+3x = 8x$ (1); Sum $= 8x+5y$ (1).	1+1+1
31	For parallel lines these angles must be equal (1); Here 65° and 65° are equal (1); The lines are parallel (1).	1+1+1
	OR: $57+75 = 132$; $132+231 = 363$ (1); Number of steps $= 2$ (1); Palindrome: 363 (1).	1+1+1
32	Two sides and the included angle are equal, so SAS; $\triangle ABC \cong \triangle PQR$, hence $PR = AC = 9$ cm (1); Perimeter of PQR $= 5+7+9$ (1); $= 21$ cm (1).	1+1+1
	OR: Improper form $\frac{5}{2}$; $\frac{5}{2} \times \frac{4}{3}$ (1); $= \frac{20}{6}$ (1); Answer $3\frac{1}{3}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Brackets (1); Multiplication and division from left to right (1); Addition and subtraction from left to right (1); Value $= 13$ (1); Order of operations (1).	1+1+1+1+1
34	Total $= 8.5$ km (1); Longest $= 3.25$ km, shortest $= 2.5$ km (1); Difference $= 0.75$ km (1); Remaining $= 10 - 8.5 = 1.5$ km (2).	1+1+1+2
35	Total $= 8r + 6$ (1); For $r = 12$: 102 (2); For $r = 14$: 118 (2).	1+2+2
36	Let the supplement be y; then the angle is $2y$ and $2y + y = 180$ (1); $y = 60$ (2); The angle is 120° and its supplement is 60° (2).	1+2+2
	OR: Sequence: $17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Halvings $= 9$ (1); Triple-and-add-1 steps $= 3$ (1); Total steps $= 12$ (2).	1+1+1+2
37	(i) Between 5 cm and 19 cm (1); (ii) Whole numbers 6 to 18: 13 values (1); (iii) Smallest third side 6: perimeter 25 cm (1); Largest third side 18: perimeter 37 cm (1); State the answers with units (1).	1+1+1+1+1
	OR: Area $= \frac{4}{5} \times \frac{5}{6} = \frac{2}{3}$ (1); Vegetable part $= \frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$ (1); Flower part $= \frac{2}{3} - \frac{1}{2}$ (1); $= \frac{1}{6}$ (1); State the three areas with units (1).	1+1+1+1+1
38	Annual use $= 2,400 \times 365 = 8,76,000$ litres (1); Nearest lakh: $9,00,000$ (1); In words: nine lakh (1); Check by estimation: 365 is about 400 (2).	1+1+1+2
	OR: Cost price $= 25 \times 60 + 10 \times 150$ (1); $= 3000$ (1); Profit/loss $= 3200 - 3000$ (1); $= 200$ (1); Profit (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–8) | Class 7 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 99,999	All digits are 9.
2	**(b)** $4 + 6 \times 5$	Evaluate each option.
3	**(c)** 4.4	Look at the hundredths digit.
4	**(b)** $4(x + 3)$	Use brackets to keep the order of operations.
5	**(c)** acute	Classify by the measure: acute $< 90°$, right $= 90°$, obtuse between $90°$ and $180°$, reflex above $180°$.
6	**(d)** 0 or 5	Recall the divisibility test.
7	**(c)** $50°, 60°, 70°$	The three angles must add up to $180°$.
8	**(d)** 2	Multiply by the reciprocal of the divisor.
9	**(a)** 76,54,322	Subtract.
10	**(b)** 28	Money paid minus cost.
11	**(d)** 3.75	Work out the bracket first.
12	**(b)** 110	Fare = fixed charge + rate × distance.
13	**(c)** $90°$	The angles around a point add up to $360°$.
14	**(b)** 40	Pairs around the middle cancel out.
15	**(c)** $\angle B = \angle C$	Equal angles and equal sides go together in an isosceles triangle.
16	**(c)** $4\frac{1}{2}$	Multiply the quantity by the number of batches.
17	**(a)** 400	1 lakh = 100 thousand.
18	**(d)** 29	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
19	**(d)** 0.375	Divide the numerator by the denominator.
20	**(b)** 19	Substitute the value of n.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Compare $75°$ and $75°$ (1); Equal corresponding angles, so the lines are parallel (1).	1+1
22	Known digit sum $= 13$; Total $= (13 + *)$ must be a multiple of 9 (1); $* = 5$; Check: 4527 has digit sum 18 (1).	1+1
23	$6x + 18 = 180$; $x = 27$ (1); Angles $64°$, $77°$, $39°$ (1).	1+1
24	$\frac{3}{4}$ of 48$= 36$; $\frac{2}{3} \times 36$ (1); $= 24$ kg (1).	1+1
25	34,56,789 rounds to 35,00,000; 12,34,567 rounds to 12,00,000 (1); Estimate $= 47,00,000$ (1).	1+1
26	First $= 64$; second $= 64$ (1); Yes, equal (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Increase $= 25,67,890 - 23,45,678 = 2,22,212$ (1); Estimate: $2,20,000$ (1); State the answer in words: two lakh twenty-two thousand two hundred twelve (1).	1+1+1
28	$16 \times 100 = 1600$ (1); $16 \times 2 = 32$ (1); $16 \times 98 = 1568$ (1).	1+1+1
29	Number of pieces $= 7.5 \div 0.25$ (1); $= 30$ pieces (1); Check: $30 \times 0.25 = 7.5$ (1).	1+1+1
30	Father's age $= 3s + 2$ (1); At $s = 10$: $3(10) + 2$ (1); Father's age $= 32$ years (1).	1+1+1
31	The sum is $90°$ and the number of parts is 3 (1); One part $= 30°$ (1); The angles are $30°$ and $60°$ (1).	1+1+1
	OR: Sum of three consecutive numbers $= 3\times$ middle (1); $63 \div 3 = 21$ is a whole number, so yes (1); The numbers are $20, 21, 22$ (1).	1+1+1
32	$\angle A + \angle B = 120°$; One part $= 120 \div 4 = 30°$ (1); $\angle A = 30°$, $\angle B = 90°$ (1); $\angle C = 180° - 120° = 60°$ (1).	1+1+1
	OR: Cloth for 8 shirts $= 8 \times \frac{5}{4}$; $= 10$ m (1); Shirts from 10\frac{5}{8} m $= 8\frac{1}{2}$ (1); Full shirts $= 8$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Rice $= 3.5 \times 52 = 182$ (1); Wheat $= 2 \times 40 = 80$ (1); Total $= 262$ (1); Balance $= 300 - 262 = 38$ (2).	1+1+1+2
34	Notebooks cost $15n$ and pens cost 24 (1); Total cost $= 15n + 24$ (1); At $n = 8$: 144 (1); Change $= 200 - 144 = 56$ (2).	1+1+1+2
35	Equal to 110°: the angle itself, its vertically opposite angle, its corresponding angle and its alternate interior angle, that is 4 angles (1); Every other angle forms a linear pair with one of these, so it measures 70° (4 angles) (2); Check: $110 + 70 = 180$ (2).	1+2+2
36	Reverse $= 549$ (1); Difference $= 945 - 549 = 396$ (1); $= 99 \times 4$ (1); In general $100a + c - 100c - a = 99(a - c)$ (1); So the difference is always a multiple of 99 (1).	1+1+1+1+1
	OR: $4x + 20 = 180$, so $x = 40$ (1); Angles 60°, 30°, 90° (1); Classification: right-angled (1); Exterior angles 120°, 150°, 90° (1); Their sum $= 360^\circ = 360^\circ$ (1).	1+1+1+1+1
37	First part $= \frac{1}{3} \times 90 = 30$ km (1); Remaining $= 90 - 30 = 60$ km (1); Second part $= \frac{3}{4} \times 60 = 45$ km (1); Left $= 60 - 45$ (1); $= 15$ km (1).	1+1+1+1+1
	OR: People surveyed $= \frac{5}{100} \times 1,20,00,000$ (1); $= 6,00,000$ (2); In words: six lakh (2).	1+2+2
38	Brackets (1); Multiplication and division from left to right (1); Addition and subtraction from left to right (1); Value $= 24$ (1); Order of operations (1).	1+1+1+1+1
	OR: Total $= 7.35$ km (1); Longest $= 3.1$ km, shortest $= 1.8$ km (1); Difference $= 1.3$ km (1); Remaining $= 10 - 7.35 = 2.65$ km (2).	1+1+1+2

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